

Cluster-Based ETF Pairs Trading: A Comparative Study of Pair Selection Methods

STAT 486 Final Project
Presentation

April 21 2026

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Research Question / Big Picture

● What are we trying to solve?

- Can we build a profitable **ETF pairs trading** strategy using only **daily data**?
- Can we make pair search tractable through **dimension reduction** and **clustering**?
- Which pair-selection rule works better in practice:
 - **Cointegration-based**
 - **Distance-based**
 - **Covariance-based**
- We evaluate everything with a **walk-forward backtest**

Core idea: reduce the ETF search space first, then compare how different ranking rules convert clustered candidates into out-of-sample performance.

Key definitions

Pairs Trading

- Trade the relative mispricing between two closely related assets [1-3].
- Construct a spread: a linear combination of the two price series.
- Enter when the spread deviates far from its historical norm.
- Long the relatively undervalued asset and short the relatively overvalued one.
- Profit if the spread mean-reverts.

ETFs

- Exchange-Traded Funds: baskets of assets that trade like stocks.
- One ETF can represent a sector, index, bond sleeve, commodity group, or region.
- **Why ETFs for pairs trading?**
 - **Noise reduction:** Reduce firm specific risks
 - **Liquidity:** Easier to find counterparties
 - **Broad Coverage:** Few hundred ETFs give exposure to thousands of underlying equities
 - **Computationally feasible:** Working with a few hundred liquid ETFs keeps clustering and backtests possible on a single machine

We pair ETFs rather than single stocks because they are **diversified, liquid, and easier to organize** into economically meaningful neighborhoods.

Data and universe

Data source and preprocessing

- Source: Yahoo Finance via yfinance [4,5].
- Frequency: daily adjusted close prices.
- Build price and return panels for a broad ETF universe.
- Use a common sample window so all methods are compared on the same support.

Final common panel

- 313 ETFs
- 2014-05-01 to 2026-04-16
- 3008 trading days

Universe coverage

- U.S. broad equity
- Sector and style ETFs
- International developed / EM
- Fixed income
- REITs
- Commodities
- Currencies

Notebook output: the retained common-history universe contains 313 tickers over 3008 daily observations.

Why ETFs? Why daily data?

Why this setting is practical

- Daily data is free and easy to reproduce.
- ETF universes cover equities, bonds, commodities, REITs, and international markets.
- Mean-reversion at a daily horizon is slow enough to test with simple walk-forward infrastructure.
- A few hundred ETFs already span thousands of underlying exposures.

Why not search all pairs directly?

- The ETF pair universe is too large for brute-force search
- Daily return co-movement is noisy and high-dimensional
- Not every highly correlated pair is truly tradable
- We need a pipeline that is both statistically meaningful and computationally manageable

Project lens: not just whether pairs trading works, but whether a principled clustering-based workflow can make the search problem tractable.

Pipeline overview



Final rolling pipeline in the notebook uses HDBSCAN for candidate-pair generation inside each formation window.

Dimension reduction

Challenge

- 313 ETFs $\times$ 3,008 days is still a large return panel.
- Raw return distances get noisy in high dimensions.
- Rolling backtests need a compact, stable representation.

PCA solution

- Standardize returns, then project into a small factor space.
- Keep the top 10 PCs for clustering.
- Remove PC1 only as a market-mode ablation.

Why this helps

Less noise
Cleaner distances
Faster rolling clustering
More stable local neighborhoods

Raw panel

313 ETFs $\times$ 3,008 days

PCA transformation

Top 10 factor scores

Clustering input

PCs kept or
PC1 removed

PCA turns the long return panel into a compact representation for clustering.

PC1 keep vs. remove

What PC1 captures

- PC1 is the dominant common factor in ETF returns.
- It captures broad market co-movement across many funds.
- We test whether keeping it helps or masks relative-value structure.

Keep PC1

- Keeps market information in clustering
- Similar broad ETFs stay close
- May help similarity-based selectors

Remove PC1

- Highlights residual co-movement
- Can yield cleaner hedge candidates
- Reduces pure beta-driven matches

Design choice

Keep market mode

Compare both

Check results

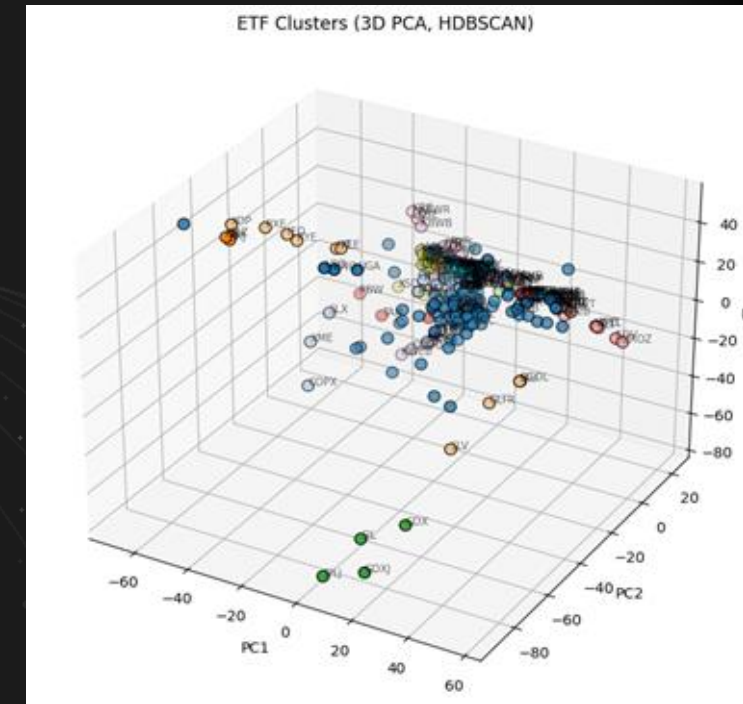
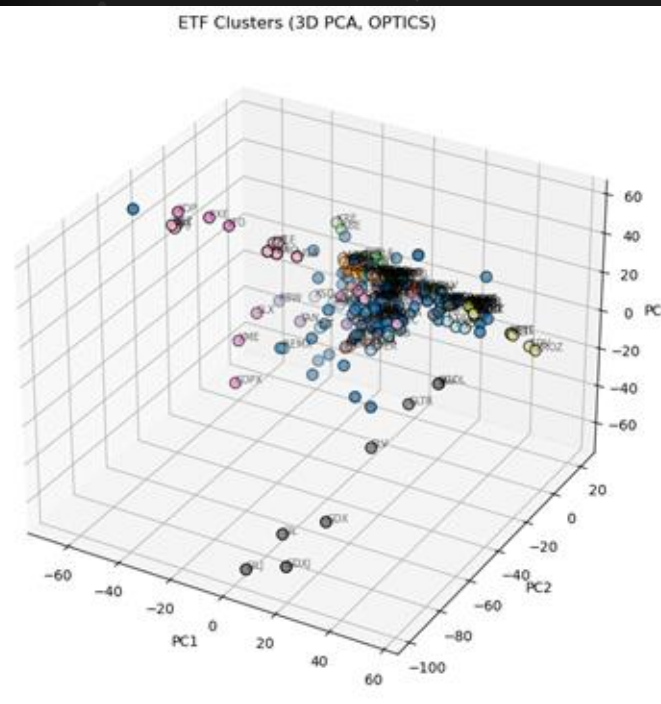
We compare both settings for all three selectors and let out-of-sample results decide.

PC1 removal is treated as an empirical test, not an assumption.

Clustering candidate pools

Why cluster first?

- Full pair search grows quickly in a 300+ ETF universe.
- Clustering limits search to local neighborhoods.
- That removes obviously unrelated cross-asset pairs.



HDBSCAN

- Builds clusters across density levels
- Marks weak points as noise
- **Why HDBSCAN?**
 - Keeps only stable, reliable clusters

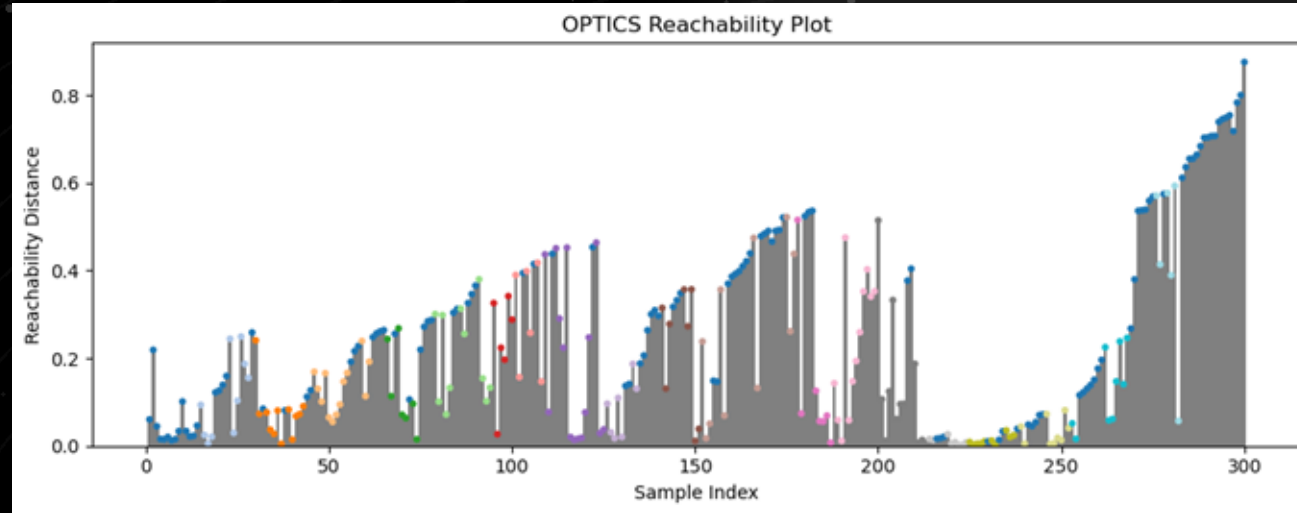
OPTICS

- Finds clusters of any shape or density
- No need to choose number of clusters
- **Why OPTICS?**
 - ETF groups have uneven densities and irregular shapes

Both methods recover sensible ETF groups; the rolling backtest uses HDBSCAN for candidate formation.

OPTICS reachability

Reachability plot from our code diagnostics



Valleys

Groups of ETFs with relatively similar return movement.

Flat local regions

Candidate neighborhoods that can support pair search.

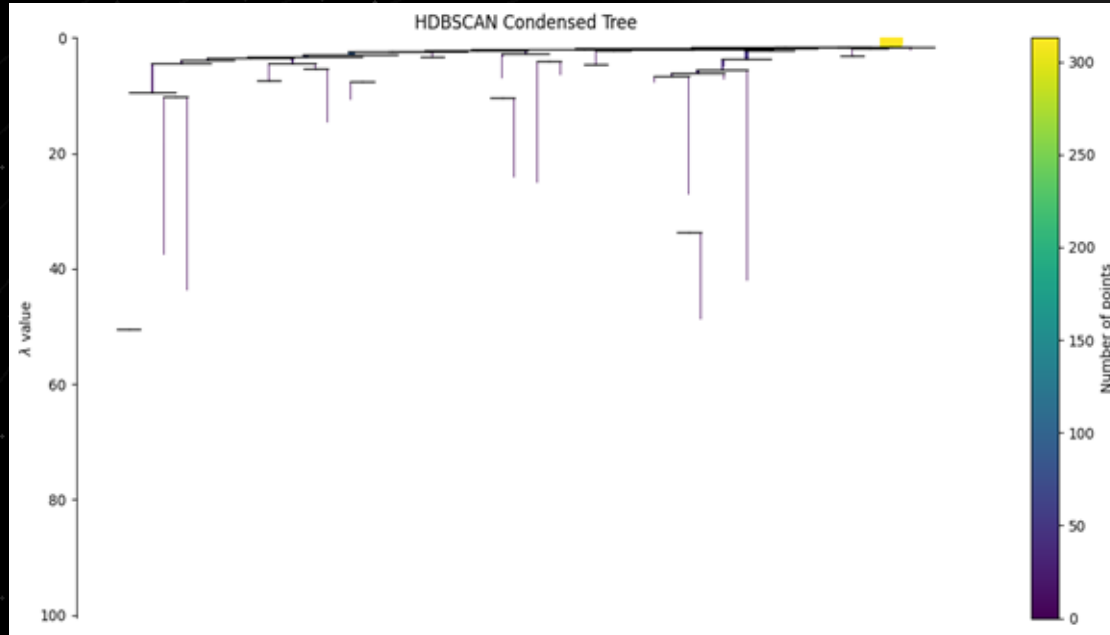
Sharp rises

Transitions between clusters or isolated noise points.

OPTICS is a diagnostic: it reveals structure in the ETF panel, but HDBSCAN gives the tighter backtest pool.

HDBSCAN condensed tree

Condensed tree from the common ETF panel



How to read it

- Long branches indicate more stable clusters.
- Short branches suggest weaker structure or noise.
- This helps separate robust groups from spurious ones.

Why it fits our pipeline

- Only within-cluster pairs move forward.
- Noise handling cuts implausible matches.
- Smaller pools make rolling selection faster.

That is why HDBSCAN becomes the formation engine in the backtest.

How clustering helps pair search

Examples from HDBSCAN neighborhoods

Type	Representative ETFs
Broad US	IVV, VTI, SCHB, SCHX, ITOT, OEF, VONE, VW
Energy	XLE, PXE, RYE, XOP, IYE, IEO
REITs	SCHH, IYR, KBWY, ICF, VNQ, RWO, RWR
Short duration FI	SHY, VGSH, NEAR, ICSH, IGSB, SPSB, SCHO, ...
Semis	SOXX, SMH, XSD

Why this matters for selection

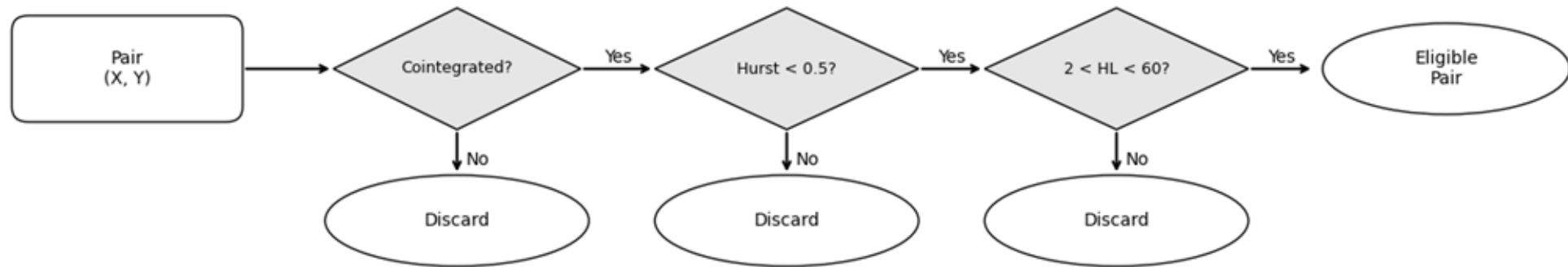
- All three selectors run after clustering.
- They compare local substitutes, not arbitrary ETFs.
- That makes rankings easier to interpret.

Practical consequence

- Fewer nonsensical pairs.
- Smaller search space in each rolling window.
- Better chance of meaningful spreads.

Clustering is the pre-filter that turns a large ETF universe into local pair-search neighborhoods.

Cointegration filter logic



Filter Definitions and Criteria

- Cointegration: Verifies the existence of a stationary linear combination between two series.
- Hurst Exponent: Assesses the degree of long-term memory; mean reversion requires $H < 0.5$.
- Half-life: Estimates the time required for a mean-reverting series to return half the distance to its mean, based on an Ornstein-Uhlenbeck (OU) model.

This filter pipeline is used only for the cointegration selector; the other selectors use the same clustered pool but different ranking rules.

Pair-selection rule 1: cointegration

How the selector works

Run Engle–Granger on price pairs.
Keep pairs with ADF p-value < 0.10 .
Require $0.20 < \text{Hurst} < 0.50$.
Require $2 < \text{OU half-life} < 60$.
Rank survivors by shorter half-life.

Pair	HL	Hurst
VEU-ACWX	2.73	0.435
PGF-PGX	2.84	0.236
SCHH-ICF	5.13	0.491
VEU-IXUS	5.34	0.444
ACWX-IXUS	5.46	0.386
JNK-HYG	5.49	0.458
XLK-IYW	9.71	0.427

Cointegration returns a small set of plausible ETF pairs, but trading performance is only moderate.

Pair-selection rule 2: distance

How the selector works

Measure formation-window return correlation.

Distance = $1 - \text{correlation}$.

Prefer the **tightest co-movers**.

Select the 10 lowest-distance pairs.

Pair	Corr	Dist
VTI-SCHB	0.9984	0.0016
MDY-IJH	0.9978	0.0022
VV-SCHX	0.9978	0.0022
IVV-VV	0.9975	0.0025
IEV-VGK	0.9974	0.0026
IVV-SCHX	0.9972	0.0028
ITOT-SCHB	0.9967	0.0033

This is the strongest rule in our tests: it repeatedly picks ETF mirrors and close substitutes.

Pair-selection rule 3: covariance

How the selector works

Compute rolling covariance in formation.

Score pairs by $\text{mean}(\text{cov}) / \text{sd}(\text{cov})$.

Require positive average covariance.

Keep only tradeable spread setups.

Select the 10 most stable pairs.

This rule is measuring **stable co-movement** strength, not stationary relative pricing

Pair	Cov Score
WEAT-CORN	7.65
PBE-XBI	7.06
XBI-FBT	6.65
IBB-XBI	5.97
PFXF-PFF	5.90
PBE-FBT	5.55
ITA-XAR	5.36

Stable covariance is intuitive, but it is weak in backtests for this ETF universe.

Walk-forward backtest design

Formation window
252 trading days



Holding window
126 trading days



Roll forward
and refit

Trading rules

Top 10 pairs each period.

Price spread: $\text{spread}_t = p_{2,t} - (\hat{\alpha} + \widehat{\beta_{\text{price}}}, p_{1,t})$

Z-score: $z_t = \frac{\text{spread}_t - \mu_{\text{spread}}}{\sigma_{\text{spread}}}$

Enter: $|z| > 2.0$; Exit: $|z| < 1.5$

Stop: $|z| > 3.0$

Signals execute with a 1-day lag.

Portfolio construction

Estimate hedge ratio and beta in formation.

Target 10% annualized spread volatility.

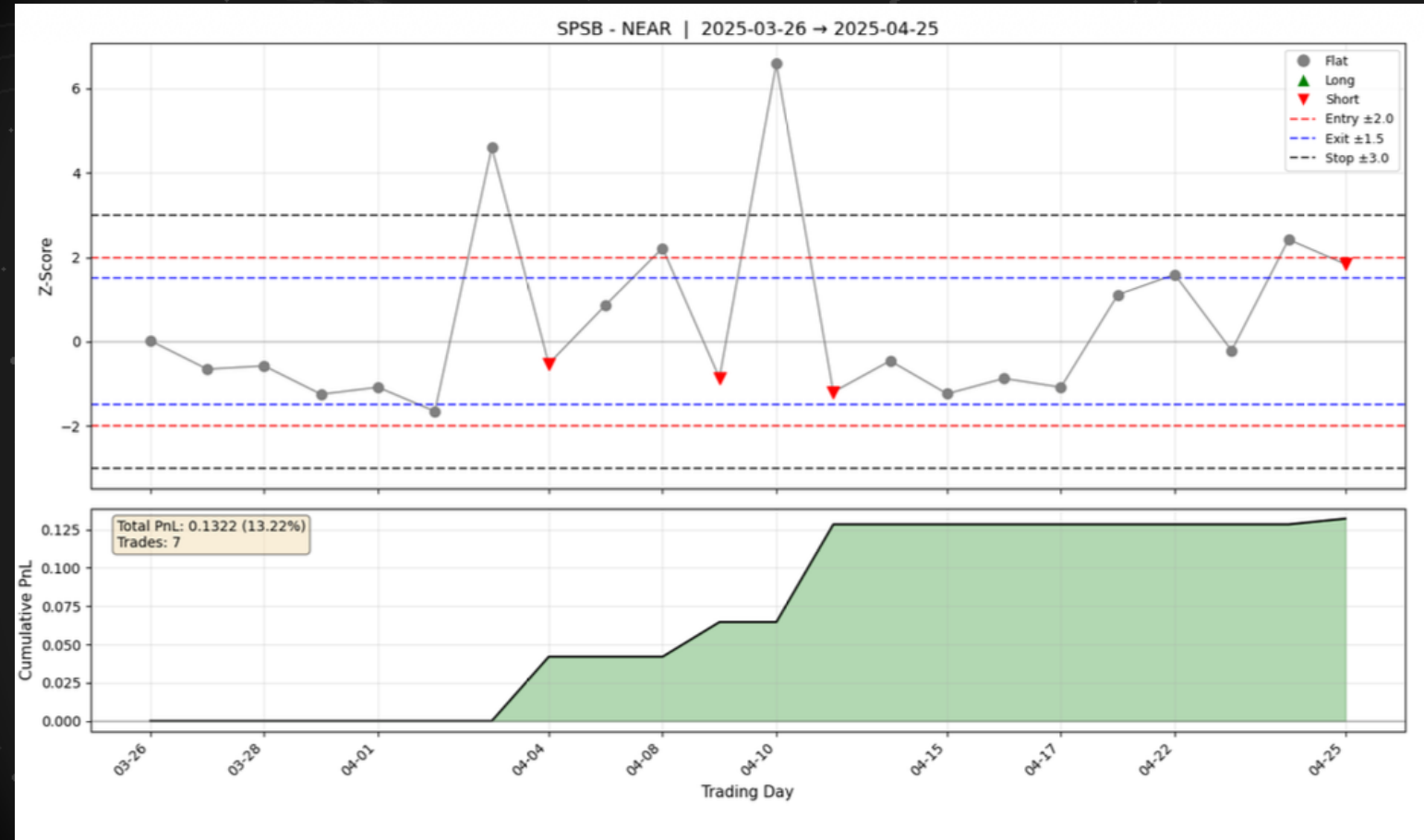
Equal-weight active pairs.

Baseline: 21 rolls, 2646 out-of-sample days.

Evaluation metrics and example trade

Evaluation metrics

- Total return: cumulative growth over the full backtest.
- Sharpe ratio: annualized mean / volatility of daily returns.
- Max drawdown: largest peak-to-trough decline.
- Win rate: fraction of positive daily pair PnL contributions in the portfolio summary used here.



Example notebook trade view (SPSB-NEAR): the upper panel shows z-score thresholds and resulting flat / long / short states, while the lower panel tracks cumulative pair PnL.

Main performance comparison

Baseline shown here matches the notebook default: HDBSCAN + 12m/6m walk-forward + PC1 removed.

Method	Sharpe↑	Max DD ↑	Win Rate↑	Final Value↑	Total Return ↑
Cointegration	0.535	-0.264	0.500	1.661	0.661
Distance	2.632	-0.131	0.552	10.533	9.533
Covariance	0.032	-0.231	0.428	0.994	-0.006

Result

Very high co-movement is more predictive than strict Engle-Granger filtering in this ETF universe.

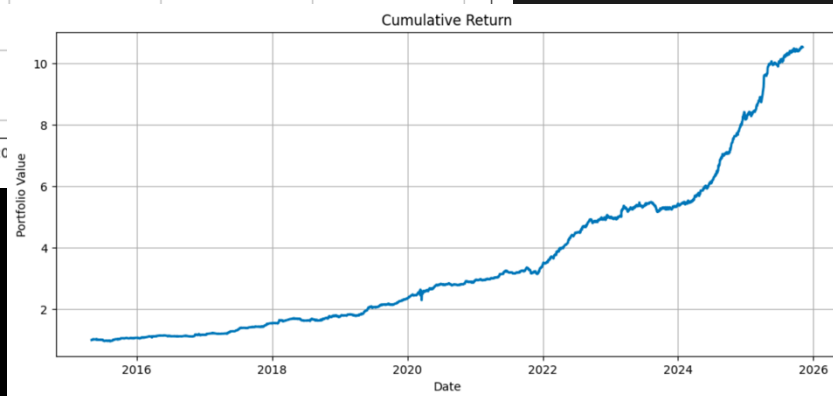
Interpretation

Very high co-movement is more predictive than strict Engle-Granger filtering in this ETF universe.

Control

Covariance is a useful comparison arm, but not a strong trading rule here.

Cumulative Return Paths Across Selection Rules

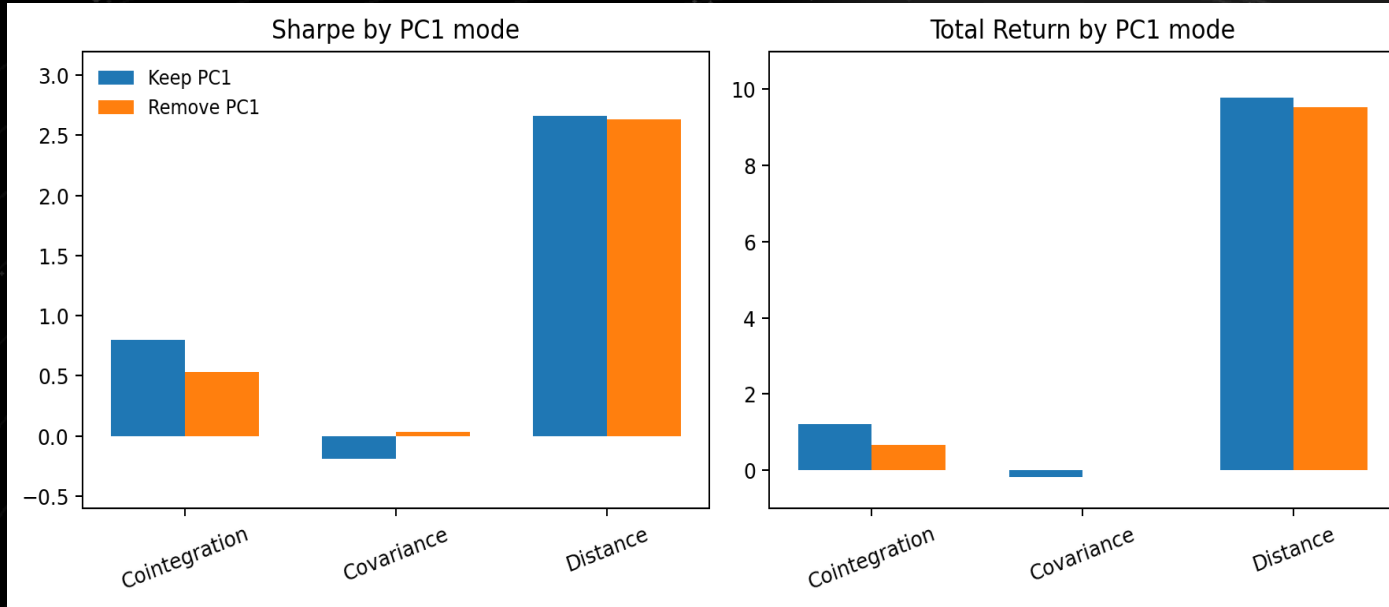


What the curves show?

- Distance compounds the fastest and most consistently over the full backtest
- Cointegration is profitable, but its growth is slower and drawdowns are more visible
- Covariance shows weak and unstable performance, ending close to flat
- These plots visually confirm the same ranking seen in the summary table: Distance > Cointegration > Covariance

Baseline shown here is the original three-way comparison: 12m formation / 6m holding with PC1 removed.

PC1 keep vs. remove



Takeaways

- Distance barely changes across the two PC1 modes.
- Cointegration improves noticeably when PC1 is kept.
- Covariance remains weak in both settings.
- So PC1 removal is not universally beneficial; its value is method-dependent.

Method	Keep Sharpe ↑	Remove Sharpe ↑	Keep Total Ret ↑	Remove Total Ret ↑
Cointegration	0.796	0.535	1.218	0.661
Covariance	-0.187	0.032	-0.177	-0.006
Distance	2.662	2.632	9.787	9.533

Statistical significance under null baselines

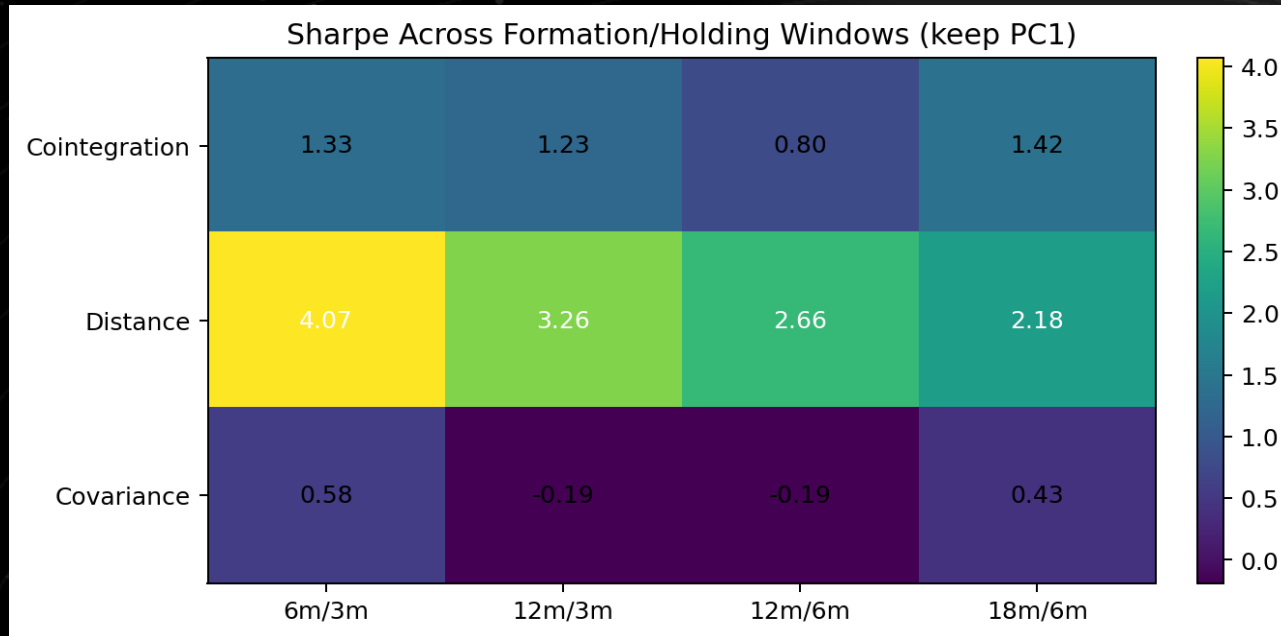
Bootstrap / permutation diagnostics use centered block resampling to ask whether observed performance materially exceeds a zero-alpha null.

PC1 Mode	Method	Obs Sharpe	Obs Tot Ret ↑	p_boot_sharpe↓
keep	Cointegration	0.796	1.218	0.004
remove	Cointegration	0.535	0.661	0.026
keep	Covariance	-0.187	-0.177	0.703
remove	Covariance	0.032	-0.006	0.413
keep	Distance	2.662	9.787	0.002
remove	Distance	2.632	9.533	0.002

Interpretation

- Distance is highly significant in both PC1 modes (bootstrap $p \approx 0.002$).
- Cointegration is also significant, though less strongly.
- Covariance is not significant and behaves like noise under the null.
- This strengthens the case that the best-performing methods are not artifacts of one lucky sample path.

Sensitivity to formation / holding windows



Heatmap shows Sharpe only. For readability, we display the keep-PC1 case here; the same overall ranking also appears under remove-PC1.

What changes across windows

- Distance is strongest in every window configuration tested.
- Best overall setting: 6m/3m with Sharpe 4.07 and total return 62.46.
- Cointegration stays positive in all tested windows, with best Sharpe 1.42 at 18m/6m.
- Covariance becomes positive in some windows, but still lags the other selectors.

Conclusion

Key Findings

- Clustering is useful because it shrinks an otherwise huge pair-search space into local ETF neighborhoods.
- Within those neighborhoods, distance-based ranking is the clear winner in the current project.
- Cointegration remains a valid and statistically significant selector, but it is more conservative and less profitable here.
- Covariance stability on its own is not enough to produce strong ETF pairs trades in this universe.
- PC1 removal is a design choice rather than a universal improvement; the best choice depends on the selector.

Headline: in this project, selection rule matters more than clustering flavor, and the strongest rule is simple distance-based similarity within clustered ETF pools.

Limitations and next steps

Current limitations

- No transaction costs, borrow fees, or slippage.
- Backtest uses daily close-to-close execution only.
- The ETF universe is broad, but not exhaustive.

Next steps

- Add realistic trading frictions and borrowing constraints.
- Test regime-aware or adaptive thresholds.
- Expand the asset universe and examine sub-universe robustness.
- Study why distance is so strong relative to stricter mean-reversion filters.

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Thank You
Questions?

Cluster-Based ETF Pairs Trading
STAT 486 Final Project

HDBSCAN

- ❑ A hierarchical density-based clustering method.
- ❑ Does not require you to specify the number of clusters in advance.
- ❑ Can mark points as noise rather than forcing every ETF into a cluster.
- ❑ In the current walk-forward code, HDBSCAN is the actual clustering engine used for candidate generation.

OPTICS

- ❑ Another density-based method that orders points to reveal clustering structure across scales.
- ❑ Its reachability plot is very useful diagnostically: valleys suggest dense clusters, sharp jumps suggest transitions or isolated points.
- ❑ In the current notebook, OPTICS is most helpful as an exploratory diagnostic and visual explanation of the clustering idea.

Appendix: Backtest Mechanics and Trading Rule

This slide summarizes the walk-forward design, trading logic, signal construction, hedge sizing, pair return formula, and portfolio aggregation used in the notebook.

1) Walk-forward design

- Formation window = 252 trading days (~12 months); holding window = 126 trading days (~6 months).
- At each rebalance date, only the formation window is used to cluster ETFs, estimate spread parameters, rank candidate pairs, and select the top 10 pairs.
- The next holding window trades those selected pairs out-of-sample; then the window rolls forward and the process repeats.

2) Signal construction

Price spread: $\text{spread}_t = p_{2,t} - (\alpha + \beta_{\text{price}} \cdot p_{1,t})$

Z-score: $z_t = (\text{spread}_t - \mu_{\text{spread}}) / \sigma_{\text{spread}}$

Symbols:

$p_{1,t}$ and $p_{2,t}$ = prices of ETF₁ and ETF₂ on day t

α = intercept from the formation-window price regression

β_{price} = price hedge ratio; μ_{spread} and σ_{spread} = formation-window mean and standard deviation of the spread

- Interpretation: $z_t > 0$ means ETF₂ is rich relative to ETF₁; $z_t < 0$ means ETF₂ is cheap relative to ETF₁.

3) Trading rule

- Use the previous day's z-score z_{t-1} to avoid same-day look-ahead bias.
- Enter long-spread if $z_{t-1} \leq -2.0$; enter short-spread if $z_{t-1} \geq +2.0$.
- Exit if $|z_{t-1}| \leq 1.5$.
- Stop-loss if the position moves further against us: for a long-spread, stop when $z_{t-1} \leq -3.0$; for a short-spread, stop when $z_{t-1} \geq +3.0$.
- For pair (ETF₁, ETF₂): long-spread = long ETF₂ and short ETF₁; short-spread = short ETF₂ and long ETF₁.

4) Return-based hedge sizing

Return regression in the formation window: $r_{2,t} = a + \beta_{\text{ret}} \cdot r_{1,t} + \varepsilon_t$

Weights: $w_1 = |\beta_{\text{ret}}| / (|\beta_{\text{ret}}| + 1)$, $w_2 = 1 / (|\beta_{\text{ret}}| + 1)$

Symbols:

$r_{1,t}$ and $r_{2,t}$ = daily returns of ETF₁ and ETF₂ on day t

a = regression intercept; β_{ret} = return hedge ratio; ε_t = residual

w_1 and w_2 = normalized position sizes for the two legs; $w_1 + w_2 = 1$

- The signal decides direction; the weights only decide the relative size of each leg.

5) Pair return calculation

Raw spread return: $r^{\text{spread}}_t = w_2 r_{2,t} - w_1 r_{1,t}$

Risk scaling: $r^{\sim\text{spread}}_t = (r^{\text{spread}}_t / \sigma_{\text{spread,ret}}) \times \text{target scale}$

Realized pair return: $r^{\text{pair}}_t = \text{pos}_t \cdot r^{\sim\text{spread}}_t$

- $\sigma_{\text{spread,ret}}$ = formation-window standard deviation of spread returns.
- target scale targets roughly 10% annualized volatility; $\text{pos}_t \in \{-1, 0, 1\}$ indicates short-spread, flat, or long-spread.

6) Portfolio aggregation and metrics

Portfolio return: $r^{\text{port}}_t = (1 / N_t) \sum_{j=1}^{N_t} r^{\text{pair}}_{j,t}$

• r^{port}_t = portfolio daily return; $r^{\text{pair}}_{j,t}$ = return of active pair j on day t ; N_t = number of active pairs on day t .

• Important: the final return is not the average return of all ETFs; it is the equal-weight average across active pair returns only.

• Reported metrics: Sharpe ratio, total return / final value, maximum drawdown, and win rate.

Key takeaway: each rebalance uses only past data to select pairs and estimate parameters. Trading in the next holding window is fully out-of-sample, positions are opened and closed using lagged spread z-scores, and the portfolio return is the equal-weight average across currently active pairs.

Why distance-based selection likely wins here

- ETFs are designed products with many close substitutes. Once clustering has already restricted the search to economically similar funds, very high return similarity is an extremely strong signal.
- Distance-based selection is less brittle than multi-filter cointegration screening. It keeps more plausible substitutes in play instead of discarding them on noisy finite-sample tests.
- In an ETF universe, close exposure matching may matter more than strict price-level cointegration tests.

Why cointegration is still respectable

- Cointegration is closer to the textbook notion of a stationary spread.
- It performs positively and significantly in multiple settings.
- Its weaker performance may reflect finite-sample test noise, strict filtering, or the fact that some very tradeable ETF substitutes do not pass all the spread-process filters cleanly.

Why covariance struggles

- Stable positive covariance means assets move together; it does not mean their relative price is stationary.
- You can have a pair with stable covariance and still have a drifting spread.
- So covariance is a good co-movement measure, but not a sufficient mean-reversion measure.